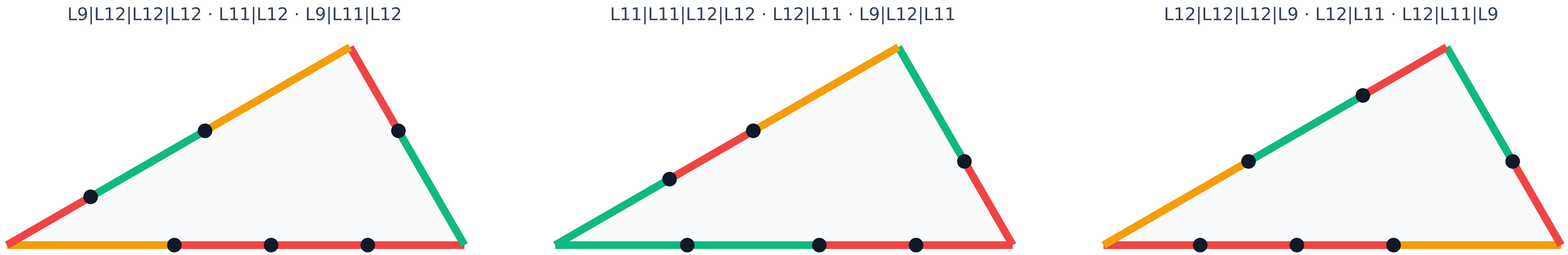
Fifteen planigons, 1,853 tiles

the same shapes, counted once per way of cutting their edges

A planigon's edge runs between the centroids of two regular polygons, so its length is the sum of two apothems, and seven of the twelve lengths that produces are themselves sums of others. Wherever that happens, one tile's edge can be met by TWO neighbours instead of one, and the atlas was not counting those tilings.

Modelling it costs, because the search glues half-edge TYPE against half-edge type: it has to be told in advance where a neighbour may start. So every edge is cut into ATOMS, the five lengths that are not sums, with a division mark at each cut. An edge with several ways to be cut is several tiles as far as the search is concerned, and a tile's count is the product over its edges.

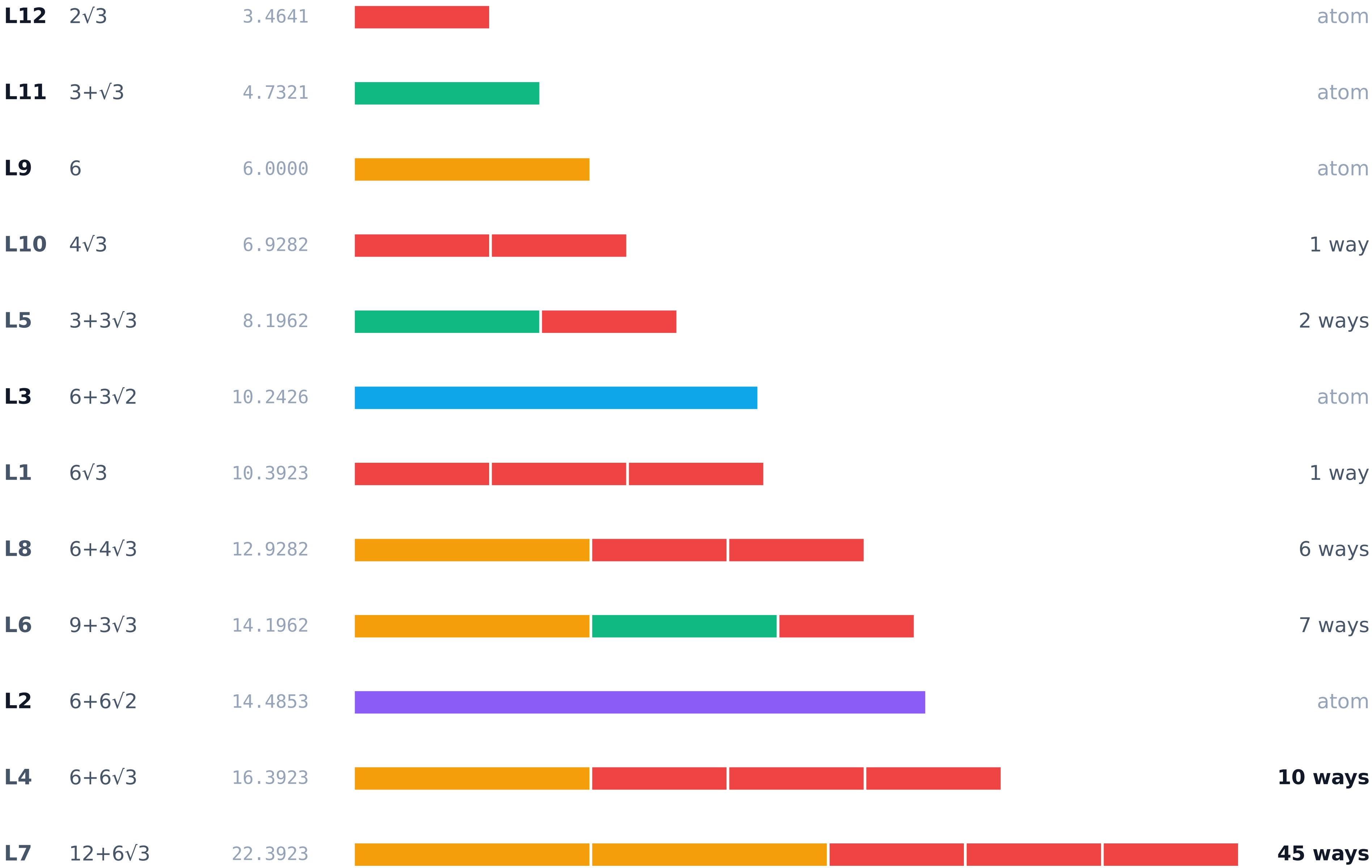
Below: one planigon, P12.6.4, drawn three times. Same 30-60-90 triangle every time. What moves is where the marks fall, and that is the whole of the difference.



one shape, three of its 140 variants

The twelve lengths, and how many ways each one cuts

atoms are drawn as one solid bar: L2, L3, L9, L11, L12 are not sums of anything



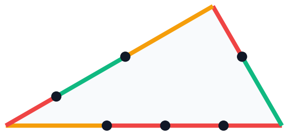
Seven lengths decompose, in 33 ways in all. L7 alone has 45 ordered atomic cuts: it is the longest edge on the palette, and it belongs to P12.12.3.

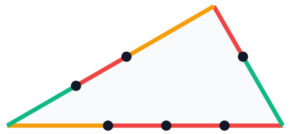
All 45 ways to cut one edge: L7 = 12+6√3

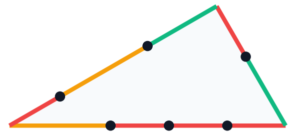
each row is one tile as far as the search is concerned

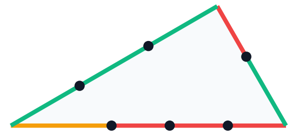
P12.6.4: all 140 variants   1 of 4

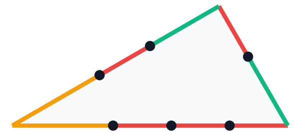
edges L4 (10 cuts) · L5 (2 cuts) · L6 (7 cuts) = 140 tiles, one shape

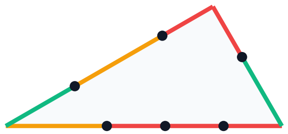

1

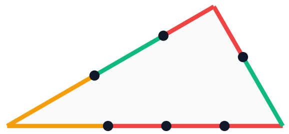

2

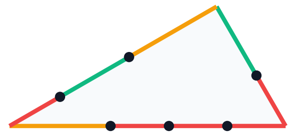

3

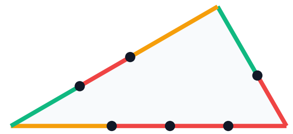

4

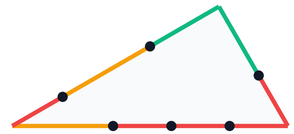

5

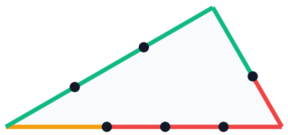

6

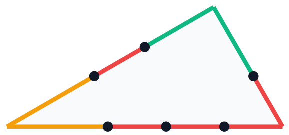

7

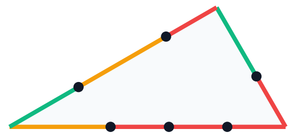

8


9

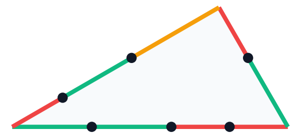

10

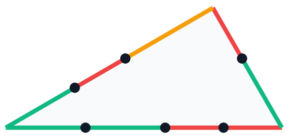

11


12

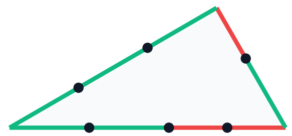

13


14

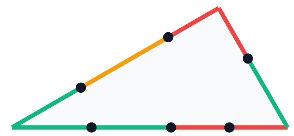

15

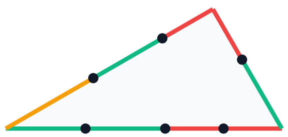

16


17

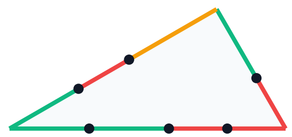

18


19

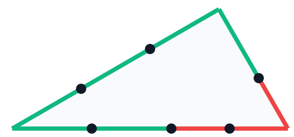

20


21

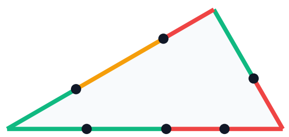

22


23


24

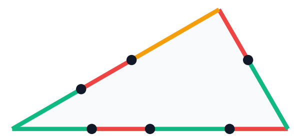

25

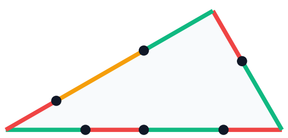

26

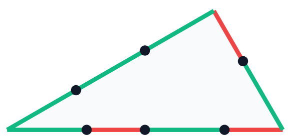

27

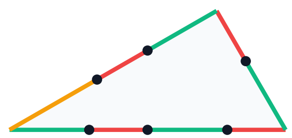

28


29

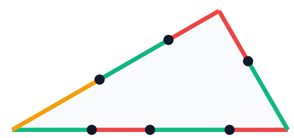

30

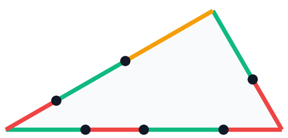

31

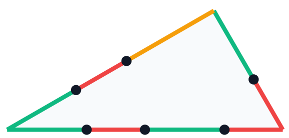

32

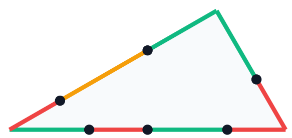

33

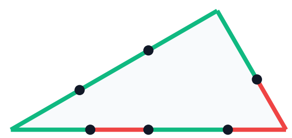

34

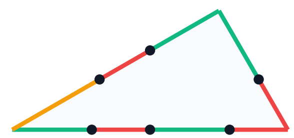

35


36


37

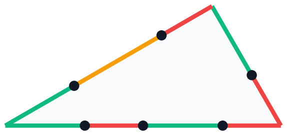
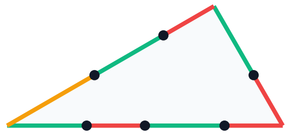
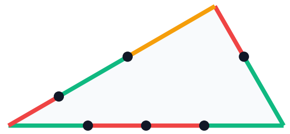
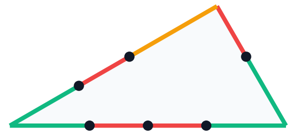
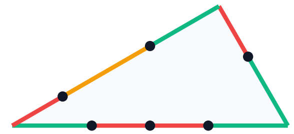
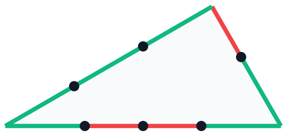
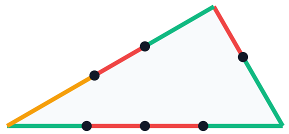
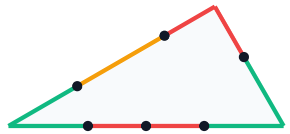
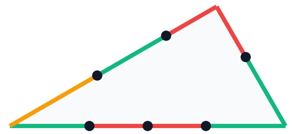
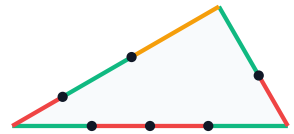
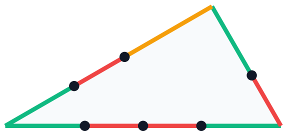

38

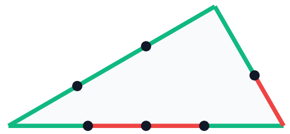
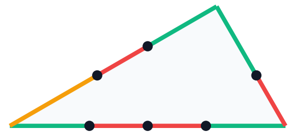
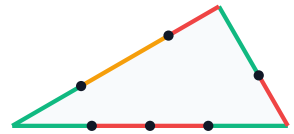
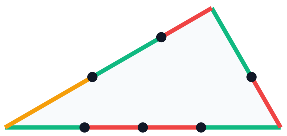
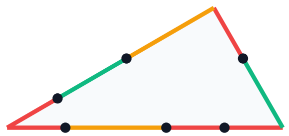
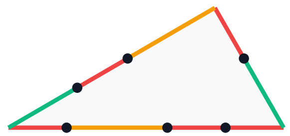
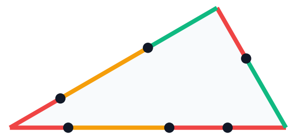
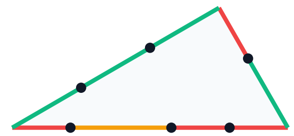
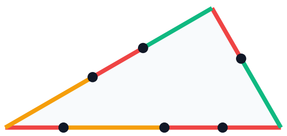
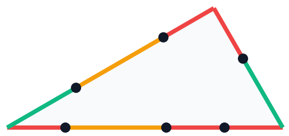
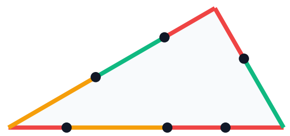
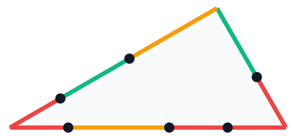
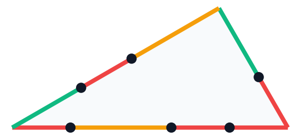
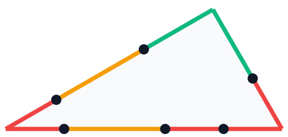
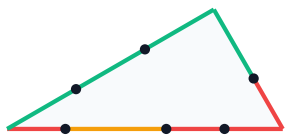
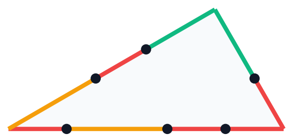
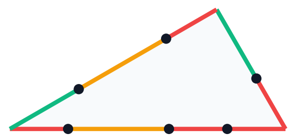
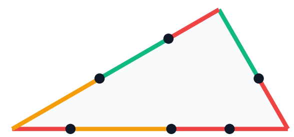
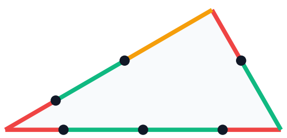
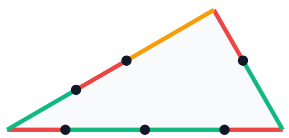
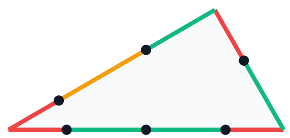
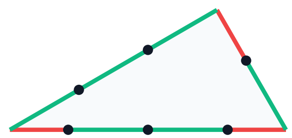
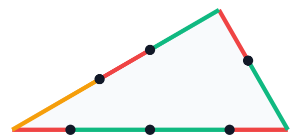
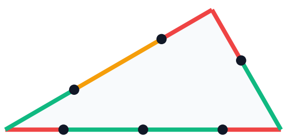
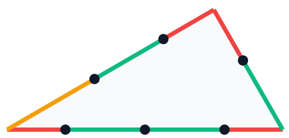
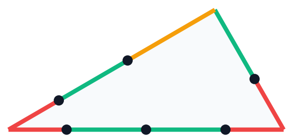
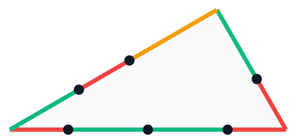
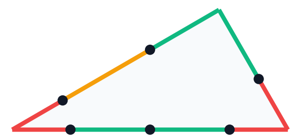

39


40

P12.6.4: all 140 variants

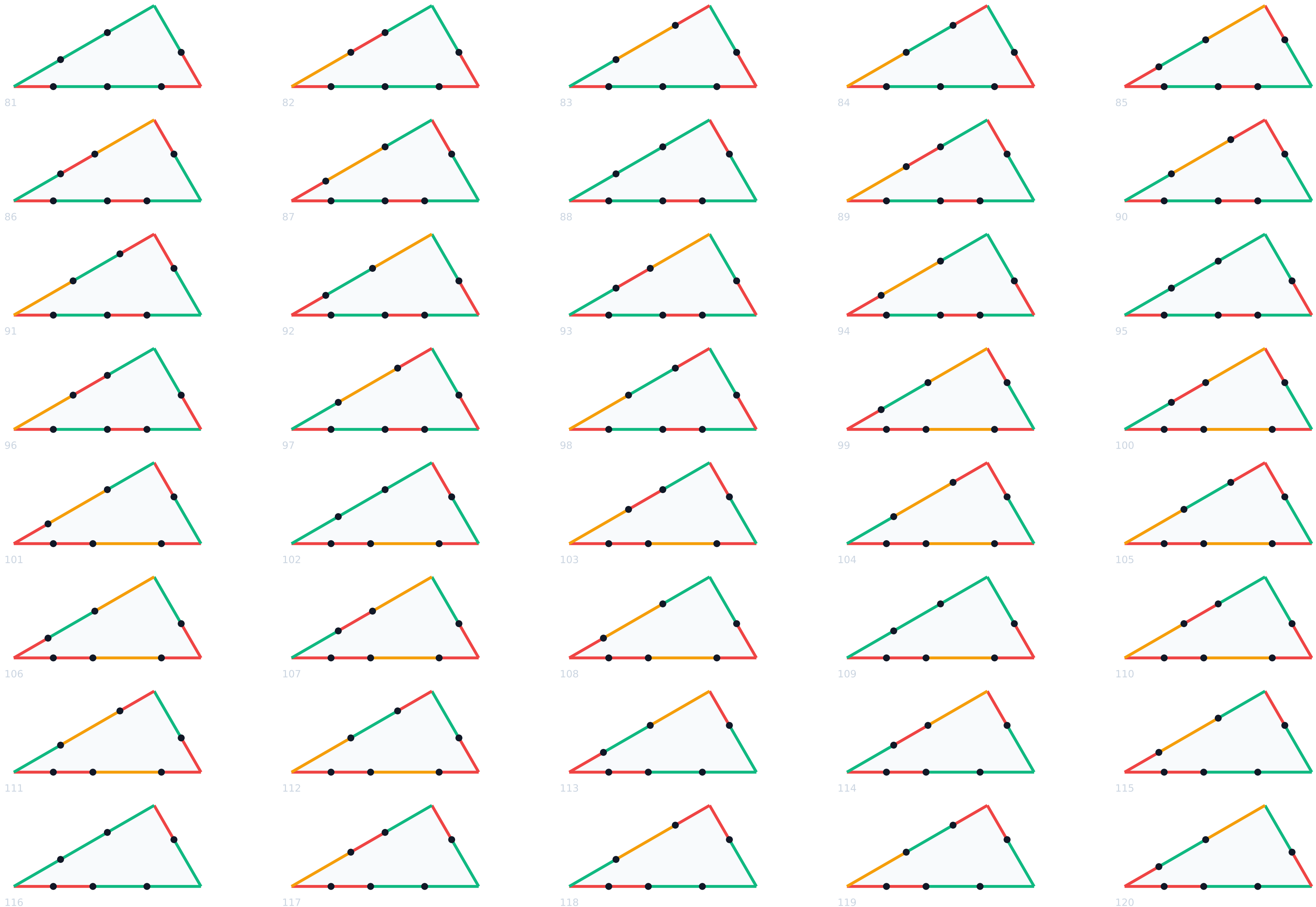
edges L4 (10 cuts) · L5 (2 cuts) · L6 (7 cuts) = 140 tiles, one shape


41

42

43

44

45

46

47

48

49

50

51

52

53

54

55

56

57

58

59

60

61

62

63

64

65

66

67

68

69

70

71

72

73

74

75

76

77

78

79

80

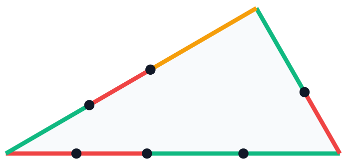
P12.6.4: all 140 variants   3 of 4

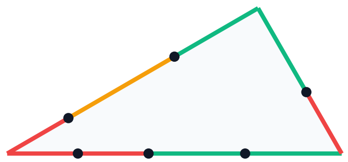
edges L4 (10 cuts) · L5 (2 cuts) · L6 (7 cuts) = 140 tiles, one shape

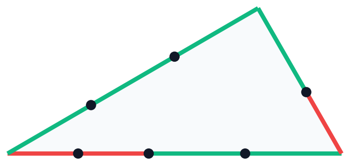


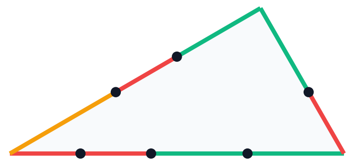
P12.6.4: all 140 variants   4 of 4

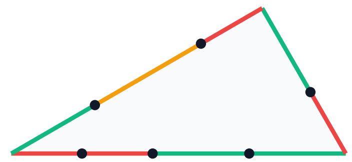
edges L4 (10 cuts) · L5 (2 cuts) · L6 (7 cuts) = 140 tiles, one shape

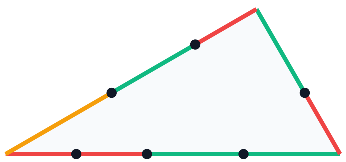

121

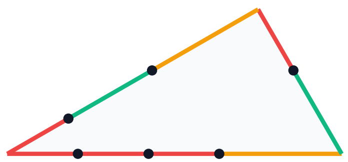

122

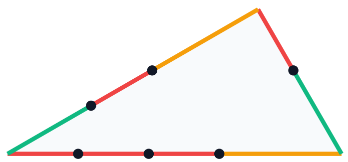

123

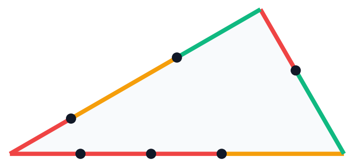

124

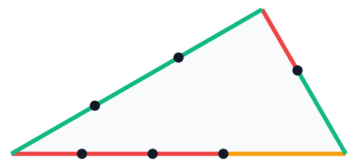

125

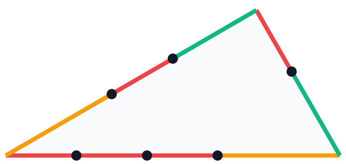

126

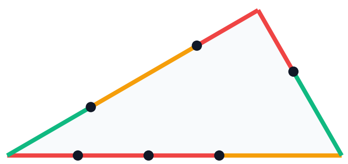

127

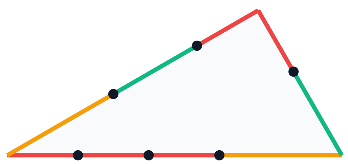

128

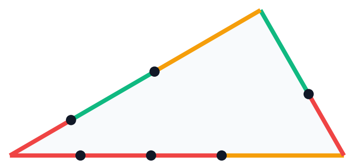

129

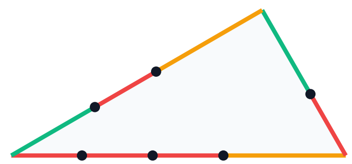

130

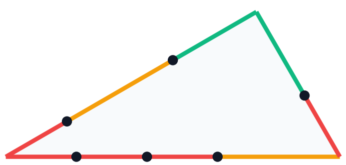

131

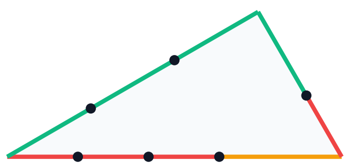

132

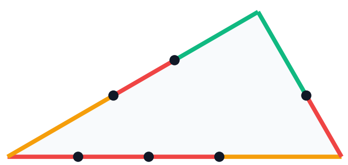

133

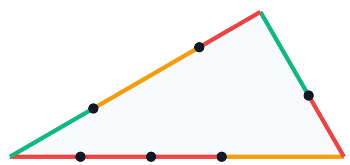

134

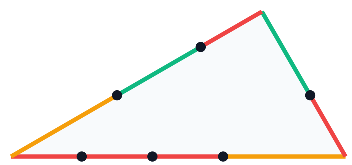

135


136


137

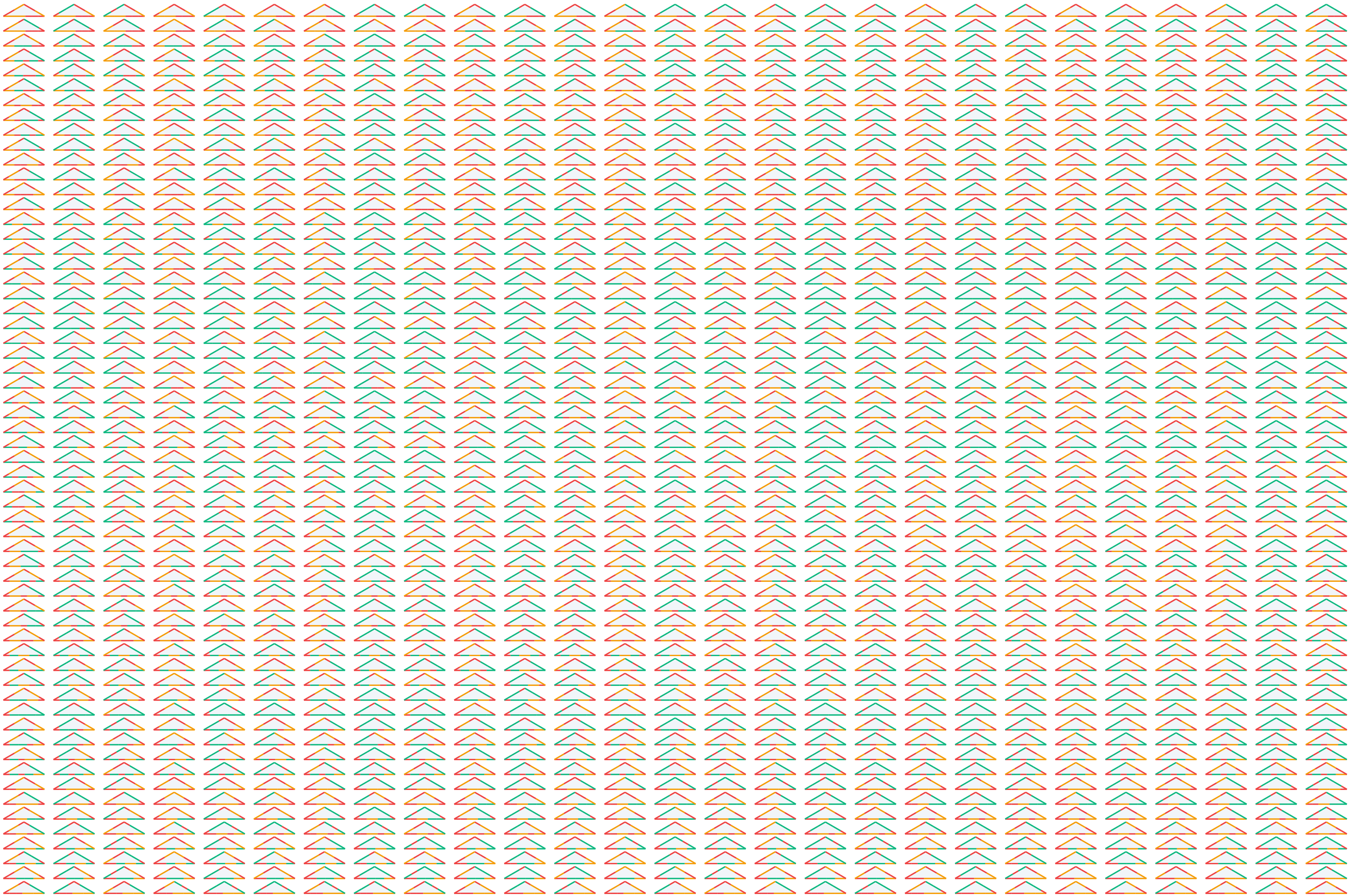

138


139


140

P12.12.3: all 1,620 variants of one triangle

edges L7 (45) · L8 (6) · L8 (6) = 1,620. Every one is the same 30-30-120 triangle. This is not a palette; it is why the run does not finish.



The whole palette

one row per planigon: its edges, the cuts each edge allows, and the product

planigon	edges	cuts per edge	variants
P12.12.3	L7 L8 L8	45 × 6 × 6	**1,620**
P12.6.4	L4 L5 L6	10 × 2 × 7	**140**
P12.4.3.3	L6 L11 L12 L8	7 × 1 × 1 × 6	**42**
P12.3.4.3	L8 L11 L11 L8	6 × 1 × 1 × 6	**36**
P6.4.3.4	L5 L11 L11 L5	2 × 1 × 1 × 2	4
P6.4.4.3	L5 L9 L11 L10	2 × 1 × 1 × 1	2
P6.6.6	L1 L1 L1	1 × 1 × 1	1
P8.8.4	L2 L3 L3	1 × 1 × 1	1
P4.4.4.4	L9 L9 L9 L9	1 × 1 × 1 × 1	1
P6.3.6.3	L10 L10 L10 L10	1 × 1 × 1 × 1	1
P6.6.3.3	L1 L10 L12 L10	1 × 1 × 1 × 1	1
P4.3.4.3.3	L11 L11 L11 L12 L11	1 × 1 × 1 × 1 × 1	1
P4.4.3.3.3	L9 L11 L12 L12 L11	1 × 1 × 1 × 1 × 1	1
P6.3.3.3.3	L10 L12 L12 L12 L10	1 × 1 × 1 × 1 × 1	1
P3.3.3.3.3.3	L12 L12 L12 L12 L12 L12	1 × 1 × 1 × 1 × 1 × 1	1
total			**1,853**

Nine planigons cost nothing: their edges are atoms already, or have exactly one atomic cut. All four expensive ones touch a DODECAGON, whose apothem is the long one, so the P12.* tiles carry the long edges and a long edge has the most ways to be cut.

The bounded run that was measured keeps the eleven cheap planigons, 15 variants in all, and drops the four P12.* tiles. On those eleven it finds 10 tilings at k=1 where matching every edge whole finds 6, and 56 at k=2 where whole edges find 27. Every one of the 4 and 29 it adds carries a T-junction, and every tiling without one was already in the whole-edge run, so the arithmetic closes exactly.

Shipping it would mean a planigon shelf missing a fifth of its tiles, so it was not shipped.